Sri Chaitanya ACADEMY | Infinity Learn



JEE CHAMPION TEST SERIES

JEE Advanced Full Syllabus Test - 5

(JEE Advanced 2021 - P2 Pattern)

Test Date: 12th March 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCE-2021-P2-Model

Time: 3:00Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 7 – 12)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 13 – 16)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 17 – 19)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 20 – 25)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 26 – 31)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 32 – 35)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 36– 38)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 39 – 44)	Questions with Multiple Correct Choice with Partial mark	+4	-2	6	24
Sec – II(Q.N : 45 – 50)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 51 – 54)	Paragraph Questions with Single Answer Type	+3	-1	4	12
Sec – IV(Q.N : 55 – 57)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

PHYSICS

Max. Marks: 60

SECTION-1(Maximum Marks: 24)

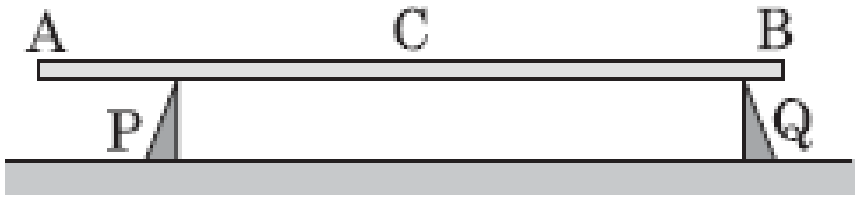
One or More Type

- This section contains SIX (06) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s)

- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If unanswered;
  Negative Marks : −2 In all other cases.

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1. A uniform bar AB is placed asymmetrically on two identical wedges P and Q that can slide on a frictionless horizontal tabletop as shown in the figure. Centre of the bar is C. Coefficient of static friction between the bar and the wedges in more than that of kinetic friction. If the wedges are shifted slowly with equal speeds on the tabletop towards each other, what will you observe?



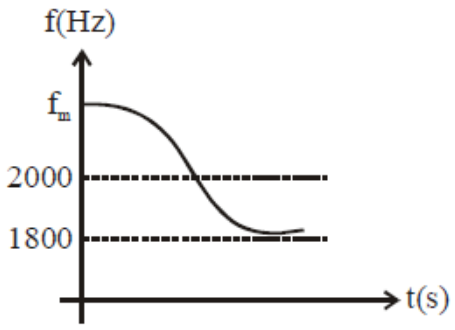
A) The wedges meet below the centre C.

B) The wedges meet somewhere between the centre C and end B.

C) The wedges always slide relative to the bar until they meet.

D) At a time, only one wedge that is farther away from the centre C slides relative to the bar.

2. A stationary observer receives a sound of frequency $f_0 = 2000Hz$ when the source is at rest. When the source starts moving with a constant velocity starting from a large distance, the apparent frequency f varies with times as shown in figure. Speed of sound = 300 m/s. Choose the **CORRECT** alternative (s)___



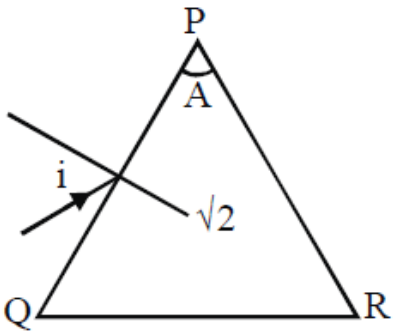
A) Speed of source is 66.7 m/s.

B) f_m shown in figure can be 2500 Hz.

C) Speed of source is 33.33 m/s.

D) f_m shown in figure cannot be greater than 2250 Hz.

3. A prism of refractive index $\sqrt{2}$ and apex angle A is shown. Light is incident from PQ side at angle of incidence $i(0 < i \leq 90°)$



A) If $A = 40°$ then light incident at all angles will emerge from surface PR.

B) If $A = 80°$ then light incident at some angle will emerge from surface PR and at some other angles light will suffer TIR at surface PR.

C) If $A = 92°$ then light incident at all angles will be reflected back from surface PR.

D) Whatever is the value of A, light will emerge from the surface PR for some value of i.

4. In terms of potential difference V, electric current I, permittivity ε_0, permeability μ_0 and speed of light c, the dimensionally correct equation(s) is (are).

A) $\mu_0 I^2 = \varepsilon_0 V^2$ **B)** $\varepsilon_0 I = \mu_0 V$

C) $I = \varepsilon_0 cV$ **D)** $\mu_0 cI = \varepsilon_0 V$

5. A moving neutron collides with stationary H-atom in ground state. As a result in excites and then de excites. The corresponding radiation fall on a surface, having work function σ. The minimum value of required kinetic energy for neutron is E_0 and possible minimum value of de Broglie wavelength of emitted photoelectrons is λ_0. If neutron hits stationary He^+ ion instead of stationary H atom, then minimum value of kinetic energy for neutron is E_1. Then

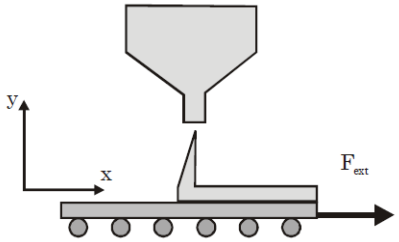
A) The value of energy transferred from neutron to H – atom is $\frac{3E_0}{4}$.

B) The value of λ_0 is $\frac{h}{\sqrt{m_e(E_0 - 2\sigma)}}$.

C) The value of E_1 is $4E_0$

D) The value of E_1 is $\frac{5E_0}{2}$

6. Sand from a stationary hopper falls onto a moving conveyor belt at a rate of 5.00 kg/s as shown in the figure. The conveyor belt is supported by frictionless rollers and moves at a constant speed of 0.750 m/s under the action of a constant horizontal external force F_{ext} supplied by the motor that drives the belt.



A) The force of friction exerted by the belt on the sand is 3.75 N.

B) The external force F_{ext} is 3.75 N.

C) The work done by F_{ext} in 1 sec is 2.81 J.

D) The kinetic energy acquired by the falling sand each second due to the change in its horizontal motion is 1.41 J.

SECTION-2(Maximum Marks: 12)
Paragraph with Numerical

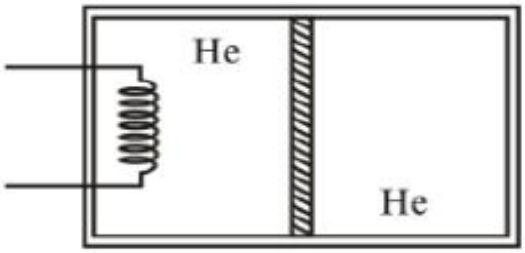
- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Question Stem for Question Nos. 7 and 8

Question Stem

A Thermally insulated cylindrical vessel is divided into two equal parts by a non-conducting piston that can move without friction. At the initial moment in the left and right parts of the vessel contains one mole of helium at the same temperature. Heat was supplied to the left side of the vessel using a heater. At the same time, the temperature of the helium in it increased by a small amount ΔT . The temperature changes by ΔT_2 on the right side of the vessel and the amount of heat Q is transferred by the heater to the system



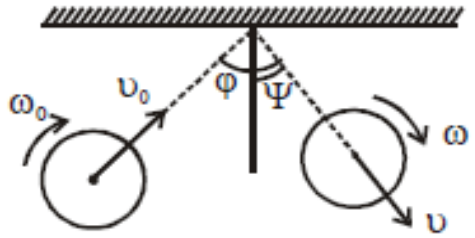
7. $\Delta T_2 = x\Delta T$. Fill x in the OMR.

8. Heat given by the heater to the system is $yR\Delta T$. Fill y in OMR sheet.

Question Stem for Question Nos. 9 and 10

Question Stem

A small solid cylindrical washer with radius $R = 4cm$ slides without friction on a horizontal ice surface with a speed of $v_0 = 10\ m/s$, while rotating around the symmetry axis with an angular velocity of $\omega_0 = 60\ rad/s$ and elastically hits a vertical wall at an angle $\varphi = 53°$ (Fig.). The friction coefficient of the washer against the wall is equal to $\mu = 0.1$.



9. The speed (in m/s) with which the washer rebounds is $\sqrt{a^2 + (b + 0.8)^2}$, then find (a + b).

10. Find the angular velocity (in rad/s) after the washer rebounds.

Question Stem for Question Nos. 11 and 12

Question Stem

The bob of mass m, charge q is revolving in a vertical circle of radius R, with the help of a massless inextensible string. If the maximum speed of the bob is V, then the period of revolution is T_1. If an electric field of magnitude (mg / q) is setup which makes an angle 60° with upward vertical. Again, the bob is circulating in same circle and its maximum speed is V, the period of revolution being T_2.

Take m= 5 kg, $g = 10\ m/s^2$, R= 2m and q = 1 C.

11. What should be the minimum value of V so that the bob is circulating in vertical circle in both the cases?

12. Find $\frac{T_1}{T_2}$?

SECTION-3(Maximum Marks: 12)

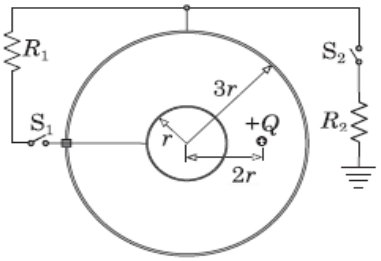
Paragraph with Single Answer Type

- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases.

Paragraph-I

Consider two electrically neutral concentric conducting thin spherical shells of radii r and 3r. A particle of positive charge Q is fixed at a distance 2r from the common centres of the shells. The inner shell can be electrically connected to the outer shell through a resistance R_1 with the help of a switch S_1 and the outer can be grounded through a resistance R_2 with the help of another switch S_2 as shown in the figure.



13. How much total heat is dissipated in the resistance R_1 after the switch S_1 is closed keeping the switch S_2 open until a steady state is reached?

A) $\frac{Q^2}{48\pi\varepsilon_0 r}$ **B)** $\frac{Q^2}{96\pi\varepsilon_0 r}$ **C)** $\frac{Q^2}{192\pi\varepsilon_0 r}$ **D)** $\frac{Q^2}{384\pi\varepsilon_0 r}$

14. When a steady state is reached after closing the switch S_1, the other switch S_2 is closed. How much total heat is dissipated after the switch S_2 is closed until a new steady state is reached?

A) Heat dissipated in each resistor is $\frac{Q^2}{48\pi\varepsilon_0 r}$.

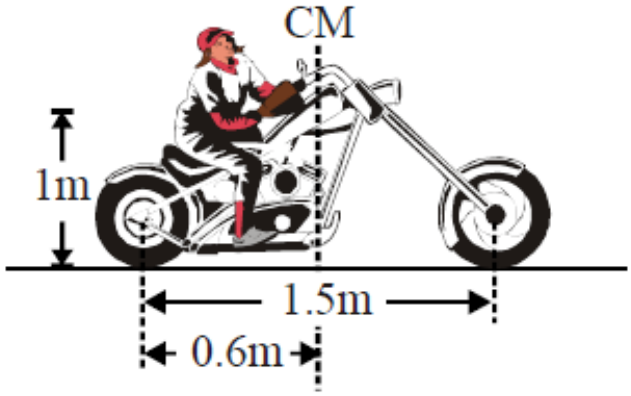
B) Heat dissipated in R_1 is $\frac{Q^2}{24\pi\varepsilon_0 r}$ and no heat is dissipated in R_1.

C) Heat dissipated in R_2 is $\frac{Q^2}{24\pi\varepsilon_0 r}$ and no heat is dissipated in R_1.

D) Both the resistors dissipate different amounts of heat, total amount of which is $\frac{Q^2}{24\pi\varepsilon_0 r}$.

Paragraph-II

There is a man of mass 100 kg sitting on a motorbike of mass 150 kg. The distance of the axles of the wheels (wheel base) is1.5m. the common mass center of the man and the motorbike is at a 1 m height above the ground level, and at a 0.6 m distance from the vertical line going through the rear axle.



15. What normal force is exerted on the rear wheel by the ground, when the motorbike starts off with an acceleration of a = 2 m/s^2 ?

A) 1000 N **B) 1250 N**

C) $\frac{2000}{3}$N **D)** $\frac{5500}{3}$N

16 What acceleration is needed to lift the front wheel ?

A) $10\ m/s^2$ **B)** $5\ m/s^2$

C) $6\ m/s^2$ **D)** None of these

SECTION-4(Maximum Marks: 12)
Non-Negative Integer Answer Type

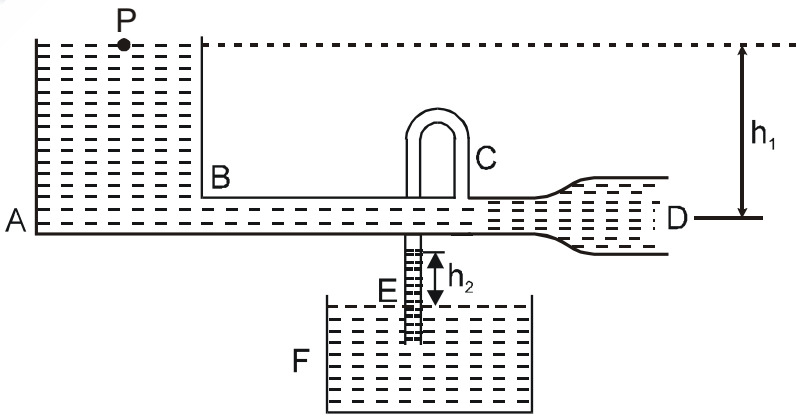
- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct integer is entered;
Zero Marks : 0 In all other cases.

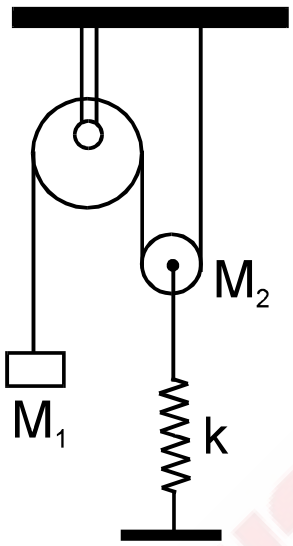
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17. Two very large open tanks A & F both contain the same liquid. A horizontal pipe BCD, having a small constriction at C, leads out of the bottom of tank A, and a vertical pipe E containing air opens into the constriction at C and dips into the liquid in tank F. Assume streamline flow and no viscosity. If the cross section area at C is one-half that at D, and if D is at distance h_1 below the level of the liquid in A, the height h_2 will liquid rise in pipe E. The relation between h_1 and h_2 given as $h_2 = kh_1$, if 'k' is positive integer, then the value of k is_____.

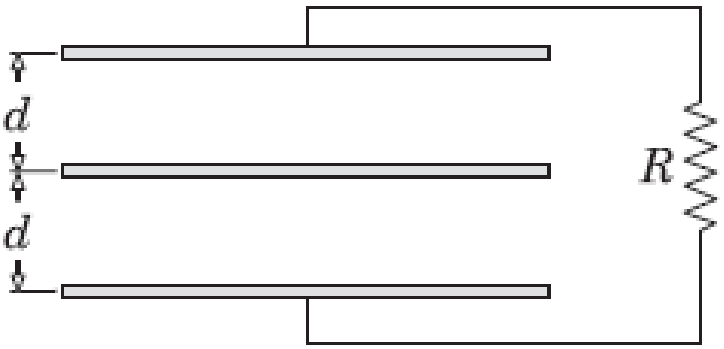
[Neglect changes in atmospheric pressure with elevation. In the containers there is atmosphere above the water surface and D is also open to atmosphere.]



18. The period of the free oscillations of the system shown here is $T = 2\pi\sqrt{\frac{M_2 + bM_1}{k}}$, if mass M_1 is pulled down a little, given that force constant of the spring is k, mass of fixed pulley is negligible and movable pulley is smooth. The value of 'b' is_____.



19. Three identical thin metal plates each of area A are arranged parallel to each other with separation d between adjacent plates and the outer plates are connected through a high resistance R as shown in the figure. A charge Q is given to the middle plate and then the middle plate is quickly shifted towards one of the outer plates by a distance d/2. The amount of heat will be dissipated in the resistor after this shift is $\frac{Q^2 d}{(4x)\varepsilon_0 A}$ (Neglect gravity and assume free space conditions). The value of '*x*' is_____.

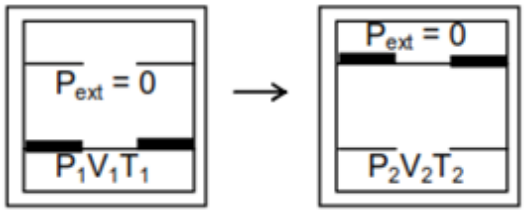


CHEMISTRY **Max. Marks: 60**

SECTION-1(Maximum Marks: 24)
One or More Type

- This section contains SIX (06) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s)
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If unanswered;
  Negative Marks : −2 In all other cases.

20. An ideal gas in thermally insulated container, at internal pressure P_1, volume V_1 and absolute temperature T_1 expands irreversibly against zero external pressure so that the pressure of gas becomes P_2, volume V_2 and temperature T_2. Which of the following statement(s) is/are correct? $(T_1 = T_2)$



A) $\Delta S_{\text{surrounding}} = 0$ **B)** $\Delta S_{\text{system}} = nR\ell n\frac{V_2}{V_1}$ **C)** $\Delta H = 0$ **D)** $P_1V_1^{\gamma} = P_2V_2^{\gamma}$

21. In an evacuated vessel of capacity 49.2 L, 4 moles of Ar(g) and 5 moles of $PCl_5(g)$ were introduced and the temperature is maintained at 240K (assume that at this temperature decomposition of PCl_5 takes place). At equilibrium, the total pressure of the mixture was found to be 4.8atm. (Given $R = 0.082 atm\ L\ mol^{-1}K^{-1}$). Correct statement(s) is/are

A) The degree of dissociation $PCl_5(g)$ into $PCl_3(g)\ \&\ Cl_2(g)$ is 0.6.

B) K_p for the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is 2.7atm.

C) At equilibrium total moles of gases are 12.

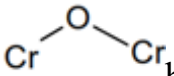
D) On removing Ar(g) from the equilibrium mixture at constant pressure and temperature, the extent of dissociation of PCl_5 will increase

22. Fusion of chromite ore $(FeCr_2O_4)$ with Na_2CO_3 in the presence of O_2 gives X which on acidification with conc. H_2SO_4 gives Y. Both X and Y contain chromium ion. Correct statement(s) is/are:

A) Both X and Y are colourful compound and colour of these compounds is due to ligand to metal charge transfer.

B) Both X and Y are paramagnetic in nature.

C) Acidified solution of Y can convert Sn^{2+} ion into Sn^{4+} ion.

D) Both X and Y have  bond.

23. Consider the following reactions (unbalanced)

$Cu + conc.HNO_3 \rightarrow K + L\uparrow + M$

$K + KCN\,(excess) \rightarrow T + S\uparrow + KNO_3$

Choose the correct option(s)

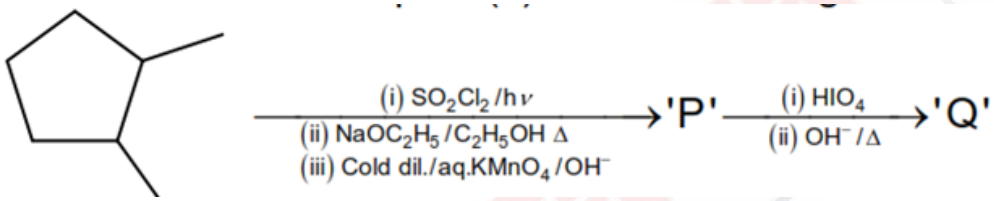
A) The oxidation state of Cu in T is +2.

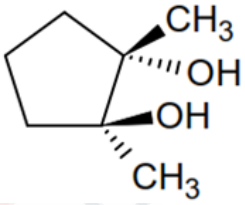
B) 'S' has 4π bonds.

C) 'L' is brown gas.

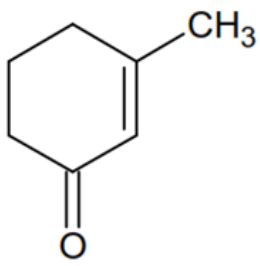
D) 'T' on reaction with H_2S does not give precipitate of CuS.

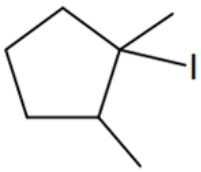
24. Choose the correct option(s) for the following reaction sequence



A) The compound 'P' is 

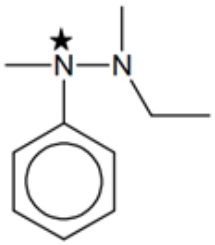
B) Total 'P' products formed are two.

C) The compound 'Q' is 

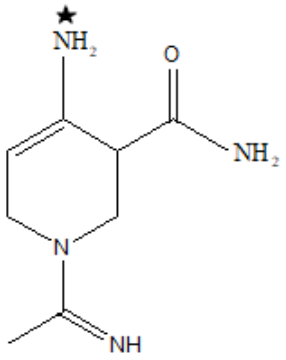
D) compound 'P' on reaction with excess of HI gives 

25. In the following compound(*) marked atom is most basic in that compound. Identify the incorrect representation.

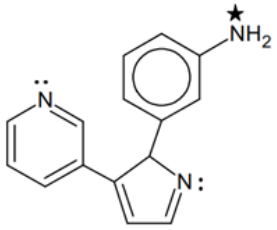
A)



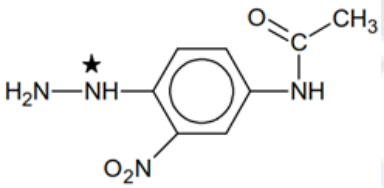
B)



C)



D)



SECTION-2(Maximum Marks: 12)

Paragraph with Numerical

- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
  Zero Marks : 0 In all other cases.

Question Stem for Question Nos. 26 and 27

Question Stem

A given 100ml aqueous solution is a mixture of 'X' g Na_2CO_3 and 'Y' g $NaHCO_3$ and 'Z'g NaCl. 20ml of solution require 20ml of $0.01M\ HCl(aq)$ to notice the end point using phenolphthalein indicator. Another 20ml of same solution require 60ml of same HCl solution to notice the end point using methylorange indicator. Given that $X+Y+Z=0.4g\left(for\ H_2CO_3\ Ka_1=4\times10^{-7}, K_{a_2}=7\times10^{-11}\ \&\log 4=0.6\ \&\ \log 7=0.8\right)$

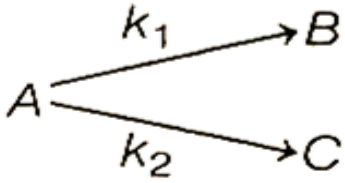
26. The weight of NaCl (the value of Z) in the given 100ml solution is

27. What is the pH of solution at phenolphalein end point

Question Stem for Question Nos. 28 and 29

Question Stem

A change to B and C in two parallel reactions,



At a time, when 80% decomposition of A takes place in 20 minutes, B and C were respectively 90% and 10%

28. Thus, $\frac{k_1}{k_2}$ is

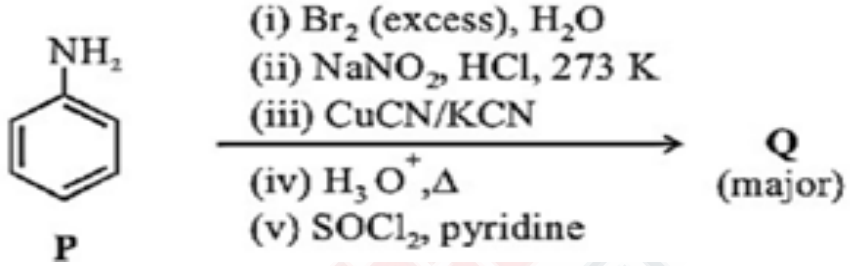
29. Rate constant k_1 is

Question Stem for Question Nos. 30 and 31

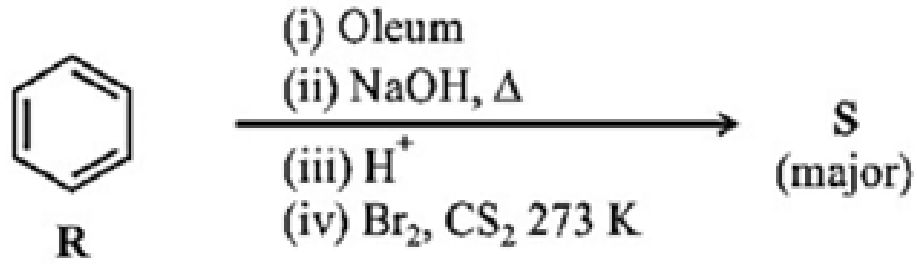
Question Stem

Scheme 1 and 2 describe the conversion of P to Q and R to S, respectively. Scheme 3 describes the synthesis of T from Q and S. The total number of Br atoms in a molecule of T is_____

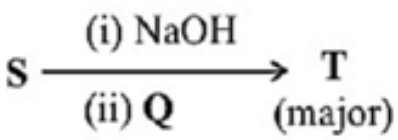
Scheme 1



Scheme 2



Scheme 3



30. Scheme 1 and 2 describe the conversion of P to Q and R to S, respectively. Scheme 3 describes the synthesis of T from Q and S. The total number of Br atoms in a molecule of T is_____

31. If compound Q is treated with NH_3/Δ and followed by reaction with $Br_2 / NaOH$ a white ppt 'U' is formed. The no of lone pairs present in the compound 'U' are _____

SECTION-3(Maximum Marks: 12)

Paragraph with Single Answer Type

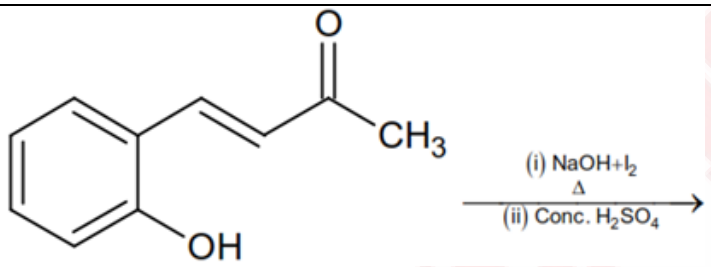
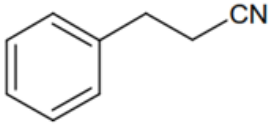
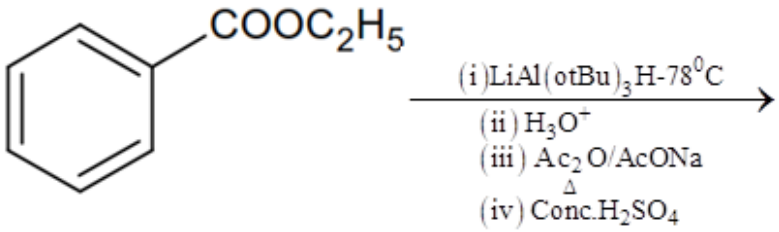
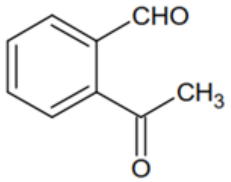
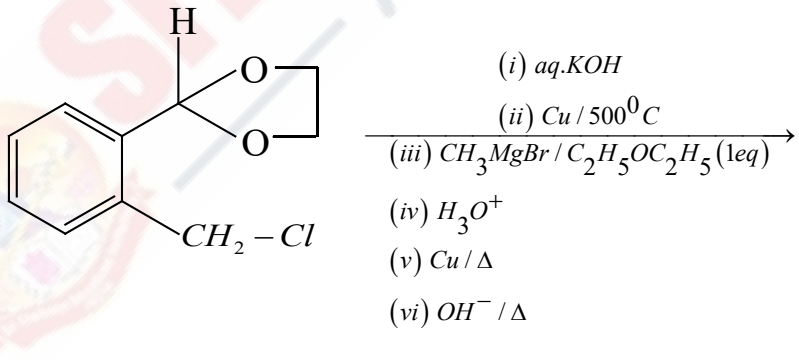
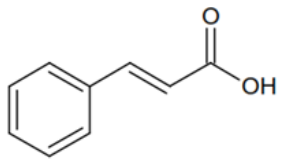
- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

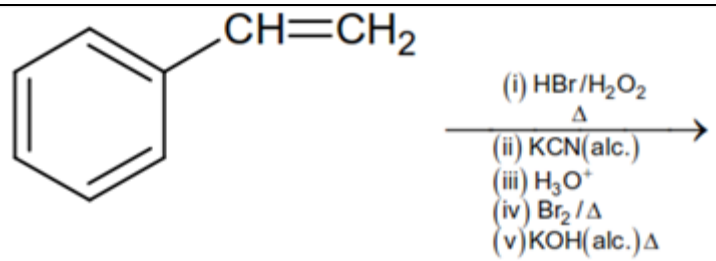
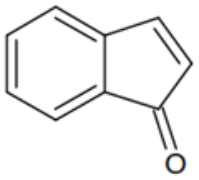
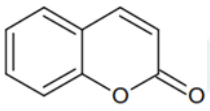
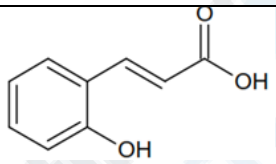
Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases.

Paragraph-1:

Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I includes starting material and reagents of selected chemical reactions. List-II gives structures of compounds that may be formed as intermediate products and / or final products from the reactions of List-I.

LIST-I		LIST-I	
(I)		(P)	
(II)		(Q)	
(III)		(R)	

(IV)		(S)	
		(T)	
		(U)	

32. Which of the following options has correct combination considering List-I & List-II?

A) $(III),(T),(U)$ **B)** $(IV),(R),(T)$

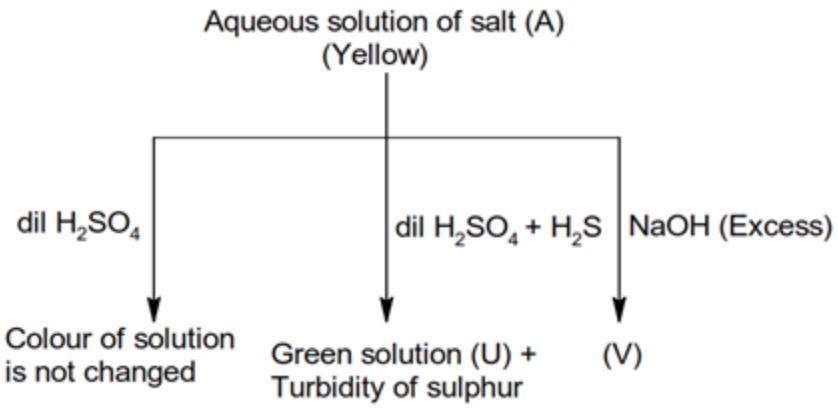
C) $(IV),(P),(S)$ **D)** $(III),(Q),(S)$

33. Which of the following options has correct combination considering List-I & List-II?

A) $(I),(U),(S)$ **B)** $(II),(R),(T)$

C) $(I),(U),(T)$ **D)** $(II),(P),(T)$

Paragraph-2:



34. Inorganic salt (A) is

A) $CrCl_3$ **B)** K_2CrO_4

C) $FeCl_3$ **D)** $AlCl_3$

35. Which observation is correct regarding (V)?

A) Brown ppt **B)** Yellow ppt

C) Brown solution **D)** Yellow solution

SECTION-4(Maximum Marks: 12)
Non-Negative Integer Answer Type

- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct integer is entered;
Zero Marks : 0 In all other cases.

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36. A current strength of 9.65 A is passed for 10 seconds through 1L of solution of 0.1 M aqueous $CuSO_4$. The pH of solution is .... (Assuming that during the process there is no change in volume).

37. The ammonia prepared by treating ammonium sulphate with NaOH is used $CuSO_4.5H_2O$ to form a stable coordination compound. If 330g of pure ammonium sulphate and 499g of pure $CuSO_4.5H_2O$ are used in preparation and $NaOH$ is used in excess, The moles of Copper – Ammonia coordination compound produced is Y. The Value of 4Y is equal to (Atomic weight in $g\,mol^{-}H=1, N=14, O=16, S=32, Na=23, Cu=63.5$)

38. In the following compounds some undergo disproportion with conc. NaOH and some do not give disproportionation with $conc.NaOH$.

$F_2, Cl_2, Br_2, NO_2, CH_3COCH_3, CCl_3CHO, P_4, I_2, (CH_3)_2\,CHCHO, HCHO.$

'x' are the number of compound which undergo dispropotionation and 'y' are the number of compound which do not undergo dispropotionation , Then value of 3x/y is ____

MATHEMATICS

Max. Marks: 60

SECTION-1(Maximum Marks: 24)
One or More Type

- This section contains SIX (06) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONE OR MORE THAN ONE of these four option(s)
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If unanswered;
  Negative Marks : −2 In all other cases.

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39. If $\sum_{r=1}^{n} T_r = n\left(2n^2+9n+13\right)$, and the sum $\sum_{r=1}^{n}\sqrt{T_r} = f(n)$ (where $f: N \to R$), then:

A) $\frac{f(100)}{\left(2\,f(50)\right)} = \frac{103}{53}$

B) $f\left(\sqrt{2}\right) = \sqrt{3}\left(\sqrt{2}+3\right)$

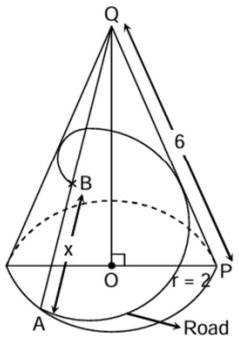
C) $\lim_{n\to\infty}\frac{f(n)}{n^2} = \sqrt{\frac{3}{2}}\left(n \in N\right)$

D) $\lim_{n\to\infty}\frac{f(n)}{n^2} = \sqrt{3}\left(n \in N\right)$

40. If $f(x) = x + \sin x$, then the value of $\int_{\pi}^{2\pi} f^{-1}(x)\,dx$ is

A) $\frac{3\pi^2}{2}-2$ **B)** $\frac{\pi^2}{2}-2$ **C)** $\frac{3\pi^2}{2}+2$ **D)** $\frac{\pi^2}{2}+2$

41. The figure illustrates a right circular cone shaped mountain. The shortest distance road is built for sight-seeing around the mountain, in which road starts at point A and ends at B(A,B and Q are collinear). Let $f(x)$ be the total length of road. The road will go up-hill for some distance $L(x)$ and down-hill for the distance of $g(x)$. Given $PQ = 6; OP = 2; AB = x$, then



A) $f(x) = \sqrt{x^2-18x+108}\,\forall x \in [0,6]$

B) Maximum value of $g(x)$ is $3\sqrt{3}$

C) $g(x) = \frac{f(x)}{\sqrt{x^2-15x+54}}\,\forall x \in [0,6]$

D) If at $x = x_0, g(x)$ takes a maximum value, then $f(x_0) = 6\sqrt{3}$

42. Let T be the line passing through the points $P(-2,7)\ and\ Q(2,-5)$. Let F_1 be the set of all pairs of circles (S_1,S_2) such that T is tangent to S_1 at P and tangent to S_2 at Q, and also such that S_1 & S_2 touch each other at point, say, M. Let E_1 be the set representing the locus of M as the pair (S_1,S_2) varies in F_1. Let the set of all straight line segments joining a pair of distinct points of E_1 and passing through the point $R(1,1)$ be F_2. Let E_2 be the set of the mid-points of the line segments in the set F_2. Then, which of the following statement(s) is (are) TRUE?

A) The point (-2,7) lies in E_1

B) The point $\left(\frac{4}{5},\frac{7}{5}\right)$ does NOT lie in E_2

C) The point $\left(\frac{1}{2},1\right)$ lies in E_2

D) The point $\left(0,\frac{3}{2}\right)$ does NOT lie in E_1

43. Let S be the set of all column matrices $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ such that $b_1,b_2,b_3 \in \mathbb{R}$ and the system of equations (in real variables) $-x+2y+5z=b_1,\ 2x-4y+3z=b_2, x-2y+2z=b_3$ has at least one solution. Then, which of the following system(s) (in real variables) has (have) at least one solution for each $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$?

A) $x+2y+3z=b_1, 4y+5z=b_2$ and $x+2y+6z=b_3$

B) $x+y+3z=b_1, 5x+2y+6z=b_2$ and $-2x-y-3z=b_3$

C) $-x+2y-5z=b_1, 2x-4y+10z=b_2$ and $x-2y+5z=b_3$

D) $x+2y+5z=b_1,\ 2x+3z=b_2$ and $x+4y-5z=b_3$

44. Let P_1 and P_2 be two planes given by

$P_1: 10x+15y+12z-60=0$,

$P_2: -2x+5y+4z-20=0$.

Which of the following straight lines can be an edge of some tetrahedron whose two faces lie on P_1 and P_2 ?

A) $\frac{x-1}{0}=\frac{y-1}{0}=\frac{z-1}{5}$

B) $\frac{x-6}{-5}=\frac{y}{2}=\frac{z}{3}$

C) $\frac{x}{-2}=\frac{y-4}{5}=\frac{z}{4}$

D) $\frac{x}{1}=\frac{y-4}{-2}=\frac{z}{3}$

SECTION-2(Maximum Marks: 12)

Paragraph with Numerical

- This section contains THREE (03) question stems.
- There are TWO (02) questions corresponding to each question stem.
- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Question Stem for Question Nos. 45 and 46

Question Stem

Consider three sets $E_1=\{1,2,3\}, F_1=\{1,3,4\}$ and $G_1=\{2,3,4,5\}$. Two elements are chosen at random, without replacement, from the set E_1, and let S_1 denote the set of these chosen elements. Let $E_2=E_1-S_1$ *and* $F_2=F_1\cup S_1$. Now two elements are chosen at random, without replacement, from the set F_2 and let S_2 denote the set of these chosen elements. Let $G_2=G_1\cup S_2$. Finally two elements are chosen at random, without replacement, from the set G_2 and let S_3 denote the set of these chosen elements . Let $E_3=E_2\cup S_3$. Given that $E_1=E_3$.

45. Let p be the conditional probability of the event $S_1=\{1,2\}$. Then p = ?

46. Let q be the conditional probability of the event $F_2=\{1,2,3,4\}$ then 3q =?

Question Stem for Question Nos. 47 and 48

Question Stem

Let p be an odd prime number and T_p be the following set of 2×2 matrices:

$$T_p=\left\{A=\begin{bmatrix} a & b \\ c & a \end{bmatrix}: a,b,c\in\{0,1,2,...,p-1\}\right\}$$

47. The number of A in T_p such that A is either symmetric or skew-symmetric or both, and det(A) divisible by p is _____ (When $p=61$)

48. The number of A in T_p such that the trace of A is not divisible by p but det(A) is divisible by p is_____(when $p=23$)

[Note : The trace of a matrix is the sum of its diagonal entries.]

Question Stem for Question Nos. 49 and 50

Question Stem

In ΔABC, let $BC=a$, $CA=b$, $AB=c$. Let P be a point on the plane of ΔABC, and G be the centroid of ΔABC.

49. If $PA=7, PB=8, PC=6$ and $PG=5$, then $a^2+b^2+c^2=?$

50. For the above triangle, let Q be any point in the plane of ΔABC. If $QA^2+QB^2+QC^2$ takes its minimum possible value at $Q=Q_0$. Then $Q_0G^2=$

SECTION-3(Maximum Marks: 12)
Paragraph with Single Answer Type

- This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases.

Paragraph-1:

A non-zero polynomial with real coefficients has the property that $f''(x).f'(x)=f(x)$ $\forall x \in R$. Then

51. The leading coefficient of $f(x)$, is

A) 1/6 **B)** 1/9 **C)** 1/12 **D)** 1/18

52. The degree of $f(x)$, is

A) 2 **B)** 3 **C)** 4 **D)** 5

Paragraph-II:

Consider a quadratic equation $az^2+bz+c=0$, where a, b, c are non-zero complex numbers.

53. The condition that the equation has one purely imaginary root is

A) $(c\bar{a}-a\bar{c})^2=-(b\bar{c}+c\bar{b})(a\bar{b}+\bar{a}b)$ **B)** $(c\bar{a}+a\bar{c})^2=(b\bar{c}+c\bar{b})(a\bar{b}+\bar{a}b)$

C) $(c\bar{a}-a\bar{c})^2=(b\bar{c}-c\bar{b})(a\bar{b}-\bar{a}b)$ **D)** $(c\bar{a}+a\bar{c})^2=-(b\bar{c}+c\bar{b})(a\bar{b}+\bar{a}b)$

54. If equation has two purely imaginary roots, then which of the following must **NOT** be true.

A) $a\bar{b}$ is purely imaginary **B)** $b\bar{c}$ is purely imaginary

C) $c\bar{a}$ is purely real **D)** $a\bar{c}$ is purely imaginary

SECTION-4(Maximum Marks: 12)
Non-Negative Integer Answer Type

- This section contains THREE (03) questions.
- The answer to each question is a NON-NEGATIVE INTEGER.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct integer is entered;
Zero Marks : 0 In all other cases.

55. Normals are drawn to the parabola $y^2=4x$ from any point on the line y = 2, then vertices of the triangle formed by tangents at feet of perpendiculars lie on a fixed rectangular hyperbola $xy=-c^2$ where c^2 is _____

56. Let $f(a,b)=\sqrt{\left(\sqrt{25-a^2}-b\right)^2+\left(a-\frac{1}{3}\sqrt{144-16b^2}\right)^2}$. If maximum and minimum value of $f(a,b)$ is c and d respectively $\forall\, a,b\in R$, then c – d equals.

57. Consider a polynomial $f(x)=a_1x^3+a_2x^2+a_3x+a_4$ such that $a_1,a_2,a_3,a_4\in\{0,1,2,.....8,9\}$. If the number of such polynomials of degree three or less than three which satisfy $a_1-a_2+a_3-a_4+9=0$ is 'n', then $\frac{n}{10}=$

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